

Alireza Norouziadzad

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Summary

AI/ML Research Scientist specializing in deep learning, computer vision, and medical imaging, currently completing a Ph.D. dissertation on generalizable, clinically-actionable AI for medical image segmentation. Developed metaheuristic-guided channel selection frameworks and uncertainty-aware robust loss functions that achieve **state-of-the-art segmentation performance** (91.84% Dice) while cutting GPU memory by **26%** and annotation burden by **37%**. Led a systematic benchmarking study of **25 medical LLMs** across 7,300+ clinical queries. **4 filed patents** and **16+** publications in high-impact journals (**IF: up to 19.2**). Proven ability to translate research into practical solutions with industry experience.

Experience

AI & ML Expert

Sep 2025–Jul 2026

[WearDOXX](#)

Toronto, Canada

- Researching, developing, and implementing machine learning and deep learning models for healthcare applications
- Responsible for data preprocessing, model design, optimization, training, evaluation, and deploying AI solutions into production
- Collaborating with cross-functional teams to integrate AI technologies into real-world healthcare products and services

Deep Learning Research Assistant

Sep 2024–Jul 2026

[Mitacs](#)

Vancouver, British Columbia (Remote)

- Engineering and rigorous validation of AI-powered biomedical systems for clinical translation
- Designing data pipelines, curating multimodal datasets, and implementing deep learning architectures for biomedical signals and imaging
- Conducted end-to-end experimentation from hypothesis formulation to model training, benchmarking, and deployment
- Collaborating with clinicians and engineers to refine system requirements and ensure scientific rigor in research outputs

Deep Learning Associate Researcher

May 2024–Sep 2024

[Scanbo](#)

Vancouver, British Columbia (Remote)

- Contributed to innovative healthcare solutions through advanced AI technologies
- Developed AI models for diagnostic methods enhancement and patient care improvement
- Performed data analysis, model development, and Collaborated with multidisciplinary teams on cutting-edge healthcare initiatives

Graduate Research Assistant

Jan 2024–Jul 2026

[York University](#)

Toronto, Canada

- **Metaheuristic Channel Selection:** Designed a family of swarm- and evolutionary-based feature-selection frameworks (EE-PSO, GWO, DWOA, Unified Framework) for DeepLabV3+ and U-Net, improving mIoU by up to 2.8% and cutting peak GPU memory by 26%
- **Robust Loss Functions:** Developed **DNA-TLoss** and **Adaptive T-Loss**, per-pixel uncertainty-aware loss functions with active learning, reducing Hausdorff Distance by 46% and annotation burden by 37% on colonoscopy polyp segmentation
- **EASeg-Net:** Engineered an edge-aware segmentation architecture combining multi-scale Laplacian fusion, deformable convolutions, and ViT-based perceptual supervision, achieving **state-of-the-art Dice** (91.84% ACDC, 85.81% Synapse) across seven public benchmarks
- **Clinical LLM Evaluation:** Led a systematic benchmarking study of **25 medical LLMs** across 7,321 patient queries, uncovering a universal Empathy–Actionability Gap with direct regulatory implications
- **Integrated Diagnostic Suite:** Engineered an AI-driven platform for volumetric **2D/3D segmentation**, multi-class lesion classification, and **automated structured reporting**
- **Efficient 3D Reconstruction:** Developed a pipeline for 3D volume synthesis by fusing 2D planar predictions via lightweight neural networks
- **Adversarial Augmentation:** Formulated adversarial strategies to mitigate data scarcity, improving inference stability in data-limited scenarios
- **Onco-Diagnostic Platform:** Designed a wearable biondiagnostic sensor for early breast cancer detection by analyzing inflammatory biomarkers in sweat samples

Education

Doctor of Philosophy in Computer Science

Jan 2024–Jul 2026

[York University](#) GPA: A+

Toronto, Canada

Supervisor: [Prof. Razieh Salahandish](#)

Thesis: Generalizable Deep Learning Approaches for Robust Medical Analysis

Master of Science in Computer Engineering

Sep 2020–Sep 2023

[University of Tehran](#) GPA: A+

Tehran, IR

Thesis: Fuzzy Particle Swarm Method-based Algorithm for Optimal Design and Automatic Generation of Artificial Neural Network

Patents

- Salahandish, R., Haghighy, F., **Norouzi, A.**, et al. "[Geometrically Enhanced Lateral Flow Immunoassays for Quantitative Assessment of Biomarkers in Point-of-Need Applications](#)" (US & WO PATENT)
- Salahandish, R., et al. "[All-in-One Integrated Multiplex Lateral Flow Immunoassay Platform for Quantitative, AI-Assisted Point-of-Care Detection](#)" (US PATENT)
- Salahandish, R., et al. "[A Wearable Sweat-Based Diagnostic Patch for Real-Time Self-Screening of Cancer Biomarkers](#)" (US PATENT)

Publications

- **Norouziadzad, A.**, et al. "[Robust Colonoscopy Polyp Segmentation Using Dynamic-Nu T-Loss with Multi-Scale and Uncertainty-Aware Adaptation](#)" *Frontiers in Medicine* (2026, IF:3.0)
- **Norouziadzad, A.**, et al. "Translating Deep Learning Innovations into Clinical Medical Imaging Practice." *Nature Artificial Intelligence Review* (IF: 19.2) [Submitted, early 2026]
- D. Sadighbayan, **Norouziadzad, A.**, et al. "Microfluidic Platform for Rapid Passive Preparation of Complex Biological Matrices in Point-of-Care Testing." *Nature Scientific Reports* (2026) [Accepted]
- **Norouziadzad, A.**, et al. "[Unified Metaheuristic Channel Selection Framework for Medical Image Segmentation](#)" *Journal of Machine Intelligence and Data Science* (JMIDS)
- **Norouziadzad, A.**, et al. "[Adaptive Student's T-Loss with Multi-Scale Uncertainty Modeling for Reliable Polyp Detection](#)" (EECSS 2025)
- **Norouziadzad, A.**, et al. "[Optimized DeepLabV3+ for Clinical Data Analysis through Advanced Particle Swarm Optimization-Based Channel Selection](#)" *Advanced Intelligent Systems* (2025, IF:7)
- **Norouziadzad, A.**, et al. "[Grey Wolf Optimizer Enhances Adaptive Atrous Spatial Pyramid Pooling for Efficient Multi-Scale Feature Selection in Medical Image Segmentation](#)" (AMIA-VR 2025)
- **Norouziadzad, A.**, et al. "[Breast Cancer Segmentation Using a Modified U-Net with Dynamic Whale Optimization Algorithm \(DWOA\) Channel Selection](#)" (MVML 2025)

– D. Sadighbayan, **NorouziAzad, A.**, et al. "A Passive, Pump-Free Microfluidic Platform for Progressive, Size-Based Blood Cell Separation." *μTAS* 2025

– F. Rahimi, E. Haghani, **NorouziAzad, A.**, et al. "Multiplex Lateral Flow Immunoassay Platform for Quantitative Point-of-Care Detection of Steroid Hormones: Testosterone, Progesterone, and Cortisol." *μTAS* 2025

– F. Esmaeildoost, S. Rozenblat, **NorouziAzad, A.**, et al. "WEARNOVAI-N: A Wearable Microfluidic Patch for AI-Driven Sweat Biomarker Screening Toward Personalized Medicine and Self-Monitoring." *μTAS* 2025

– Haghayegh, F., Haghani, E., **NorouziAzad, A.**, et al. "[Integrating Advanced Microfluidic Lateral Flow Systems with a Finger-Prick Blood Collection Cartridge to Create an All-in-One Platform for Point-of-Care Diagnostics](#)" *Biosensors and Bioelectronics* (2024)

– Haghayegh, F., **NorouziAzad, A.**, et al. "[Revolutionary Point-of-Care Wearable Diagnostics for Early Disease Detection and Biomarker Discovery through Intelligent Technologies](#)" *Advanced Science* (2024, IF:15.1)

– Maydanchi, M., Ziaei, A., Basiri, Mi., **NorouziAzad, A.**, et al. "[Comparative Study of Decision Tree, AdaBoost, Random Forest, Naïve Bayes, KNN, and Perceptron for Heart Disease Prediction](#)" (*IEEE SoutheastCon* 2023)

– Boreiri, Z., **NorouziAzad, A.**, Ghodousian, A. "[A Convolutional Neuro-Fuzzy Network Using Fuzzy Image Segmentation for Acute Leukemia Classification](#)" (*The 27th International Computer Conference, IEEE*, 2022)

– Boreiri, Z., **NorouziAzad, A.**, Majd, N. "[Optimized Quantum Circuits in Quantum Image Processing Using Qiskit](#)" (*2022 International Conference on Machine Vision and Image Processing (MVIP)*, IEEE, 2022)

– Ghodousian, A., **NorouziAzad, A.**, Boreiri, Z. "[FRE paths for finding the minimal solutions of the linear optimization problems subjected to Einstein-product-FREs](#)" (*The 3rd International Conference on Challenges and New Solutions in Industrial Engineering, Management and Accounting*, 2022)

– Boreiri, Z., **NorouziAzad, A.** "[A Novel Consensus Protocol in Blockchain Network based on Proof of Activity Protocol and Game Theory](#)" (*The 27th International Conference on Web Research (ICWR)*, IEEE, 2022)

– Ghodousian, A., **NorouziAzad, A.**, Amiri, H. "[Log-sum-exp optimization problem subjected to Lukaszewicz fuzzy relational inequalities](#)" (*The 1st National Conference on Optimization and New Solution Methods*, 2021)

– Ghodousian, A., Amiri, H., **NorouziAzad, A.** "[On the resolution and Linear programming problems subjected by Aczel-Alsina Fuzzy relational equations](#)" (*The 14th International Conference of Iranian Operations Research Society*, 2021)

– Ghodousian, A., Amiri, H., **NorouziAzad, A.** "[A global optimal solution for the weighted power mean programming problem constrained with fuzzy relational equalities](#)" (*The 52nd Annual Iranian Mathematics Conference*, 2021)

Leadership & Academic Service

Undergraduate Research Mentor	May 2024–Sep 2026
York University	Toronto, Canada
– Mentored more than 10 senior undergraduate students on AI and medical AI research projects.	
– Provided technical guidance on deep learning, experimental design, literature review, and computer vision implementation.	
– Supervised project progress and supported students in developing research and problem-solving skills.	

Skills

Python • PyTorch • TensorFlow • Keras • OpenCV • MATLAB • C/C++ • Docker • Git • Medical Imaging • Computer Vision • Deep Learning • Neural Networks • Metaheuristic Optimization (PSO, GWO, WOA) • Vision Transformers (ViT) • Uncertainty Quantification • Active Learning • Image Segmentation • 3D Reconstruction • LLM Evaluation • Wearable Devices • Biomedical Signal Processing • Fuzzy Logic Systems

Honors & Teaching

Honors
– Ph.D. Fellowship: Awarded full funding for doctoral studies at York University covering tuition and living expenses (2024–2026)
– Mitacs Accelerate Internship: Granted two-year research internship with \$90,000 CAD funding to develop AI-driven medical technologies for clinical applications (2024–2026)
– Distinguished M.Sc. Student Award for outstanding academic achievements, holding the Second-rank position in major (University of Tehran)
– Member of The National Organization for Development of Exceptional Talents (NODET)

Teaching Experience
York University : Artificial Intelligence, Object Oriented Programming, Electrical Circuits, Discrete Mathematics for Engineers, Introduction to Artificial Intelligence
University of Tehran : Operating Systems, Big Data Systems, Data Structures, Nonlinear Optimization, Quantum Computing, Randomized Algorithms, Graph and Network Algorithms